

## Activity 9 Pre-Lab — EAS Bromination of Aniline / N-Phenylacetamide

[Your name] & [partner]  
[MM/DD/YY]

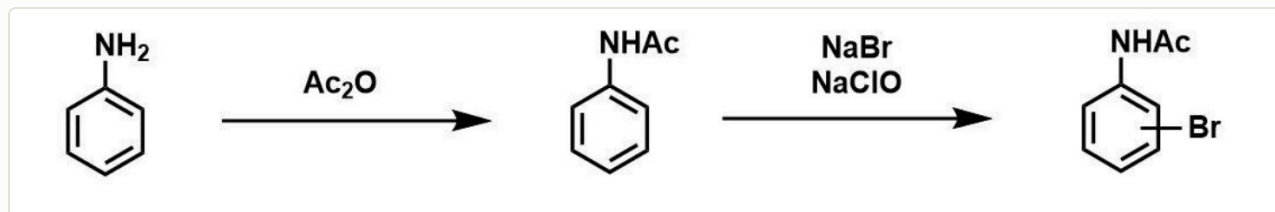
Organic Lab · electrophilic aromatic substitution

■ **Black** = write this in your notebook   ■ **Blue** = context, don't copy it   Copy the header onto every page. Draw dashed-box structures yourself.

### Purpose

Study **electrophilic aromatic substitution (EAS)**: first **protect** aniline's amine as an amide (N-phenylacetamide), then **brominate** the ring. This shows how an -NHR group is a strong **o/p-activator** and how protection tempers its reactivity so you get clean **para** product.

🔥 **DRAW** — reproduce this reaction in your notebook (from the procedure)



Step 1 protects the amine as the amide (Ac<sub>2</sub>O); Step 2 brominates — Br<sup>+</sup> is made in situ from NaBr + NaOCl (bleach) in acetic acid; the amide is an o/p-director so you get the **para** product.

### Chemical Reagents & Safety Table

**CONTEXT** ▶ Required (8): aniline, acetic anhydride, sodium hypochlorite, 95% ethanol, glacial acetic acid, sodium bromide, sodium thiosulfate, sodium hydroxide. Typical SDS values — cross-check the SDS tile. **Aniline is quite toxic** — treat carefully.

| Chemical                     | Role                                | MW     | Amount                        | Physical properties                                 | GHS pictograms |
|------------------------------|-------------------------------------|--------|-------------------------------|-----------------------------------------------------|----------------|
| Aniline                      | Starting material (limiting)        | 93.13  | ~1 mL (0.5 mL x2, ~0.011 mol) | Colorless–brown oily liquid; bp 184 °C; d 1.02 g/mL |                |
| Acetic anhydride             | Acetylating (protecting) agent      | 102.09 | 2 mL                          | Colorless liquid; bp 139 °C; d 1.08 g/mL            |                |
| Sodium hypochlorite (bleach) | Oxidant → generates Br <sup>+</sup> | 74.44  | 18 mL                         | Pale greenish aqueous liquid                        |                |
| Sodium bromide               | Bromine source                      | 102.89 | ~2.8 g                        | White crystalline solid                             |                |
| Glacial acetic acid          | Solvent / acid medium               | 60.05  | 10 mL                         | Colorless liquid; bp 118 °C; d 1.05 g/mL            |                |
| 95% Ethanol                  | Recrystallization solvent           | 46.07  | 10 mL                         | Colorless liquid; bp 78 °C; d 0.79 g/mL             |                |
| Sodium thiosulfate           | Quenches excess oxidant             | 158.11 | as directed                   | White solid (aq. soln)                              | — low hazard   |
| Sodium hydroxide (0.5 M)     | Dissolves/works up amide            | 40.00  | 3 × 10 mL                     | Colorless aqueous solution                          |                |

**GRAB FIRST:** 50-mL round-bottom + stir bar · stir plate + clamp · ice-water bath · 2× 1-mL syringes (aniline) + 25-mL Erlenmeyer + 400-mL beaker · latex gloves (aniline) then nitrile · steam bath · 125-mL Erlenmeyer · Büchner funnel + filter flask + vacuum · graduated cylinders.

### Procedure Plan (left = plan, right = fill in during lab)

#### Plan (write before lab)

##### Part 1 — Protect: aniline → acetanilide

1. Instructor adds **2 mL acetic anhydride** to a clamped **50-mL RBF** + stir bar; stopper; lower into an **ice bath** on a stir plate.
2. At the aniline hood (**latex gloves**), draw **0.5 mL aniline into each of 2 syringes** (record exact volume for theoretical yield). **Aniline is toxic + absorbs through skin — don't overdraw the syringe.**
3. Add the aniline **dropwise over 5 min** with stirring → mixture solidifies (acetanilide); break up lumps with a spatula. Clean syringes with acetone into aniline waste; switch to **nitrile gloves**.
4. Add **10 mL 0.5 M NaOH**; warm on steam bath ~1 min to loosen solid; pour into a 125-mL Erlenmeyer; rinse RBF with **2 more × 10 mL NaOH** into the same flask.
5. Cool to RT then in ice ~10 min; collect the colorless **N-phenylacetamide** by **vacuum filtration**; save an IR-size sample + take IR.

##### Part 2 — Brominate

6. Dissolve the acetanilide in **10 mL glacial acetic acid + 10 mL 95% ethanol**; add ~2.8 g **NaBr**, stir to dissolve.
7. Measure **18 mL bleach (NaOCl)** and add to the reaction; stir. **Bleach + NaBr in acid makes the Br<sup>+</sup> electrophile in situ. Never mix bleach with straight acid elsewhere — toxic gas.**
8. Quench excess oxidant with **sodium thiosulfate**, precipitate, **vacuum filter**, and **recrystallize from 95% ethanol**; weigh → % yield; mp / IR / NMR (para pattern).

#### ✍️ WRITE

aniline is limiting → theoretical acetanilide → theoretical 4-bromoacetanilide; % yield each step.

#### Experiment Notes (in lab)

Exact aniline volume: \_\_\_\_\_ mL

| step (product)           | mass | mp | % yield |
|--------------------------|------|----|---------|
| Pt 1 — N-phenylacetamide | g    | °C |         |
| Pt 2 — 4-bromo product   | g    | °C |         |

#### Part 1 — solidifying / color:

\_\_\_\_\_

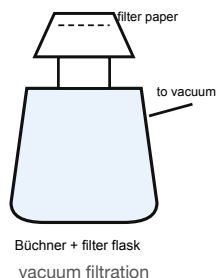
#### Part 2 — on adding bleach (color/heat):

\_\_\_\_\_

#### Para evidence (NMR: two doublets):

\_\_\_\_\_

### Setup — draw this



#### CONTEXT (don't copy) ▶

**Why protect the amine first?** free aniline is SO activated it over-brominates (tri-bromo) and its lone pair also reacts with the oxidant. Acetylating to the amide dials the activation down to a controllable **o/p-director**, so you get mono-**para** product.

**Where does Br<sup>+</sup> come from?** NaOCl (bleach) oxidizes Br<sup>-</sup> (from NaBr) in acetic acid to an electrophilic bromine species — greener than using liquid Br<sub>2</sub>.

### Clean-up & Conclusions

- Aniline rinses → labeled aniline waste; latex gloves → solid waste; filtrates → aqueous waste; organic solvents → organic waste; keep bleach away from acids; bench task.
- **Conclusions** (~4 sentences **after lab**): % yields; that the **-NHC(O)CH<sub>3</sub>** group directed bromination **para** (NMR: two doublets, para pattern); why protecting the amine was necessary; a loss source.